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DEPARTMENT OF PHYSICS

SLYT experiment

PHYSICS LABORATORY II - FUNDAMENTAL INTERACTIONS

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1 Introduction

The main goal of this project is to characterise a 64-channel LYSO scintillator. LYSO is an excellent scintillation material due to its fast response time (order 10 ns) and its high light yield. Furthermore, the high granularity of the detector (which is optically coupled to a multianode PMT) allows for spatially resolved measurements of the scintillation light.

Aside from the practical challenges of dealing with such a system (e.g. having to properly store and process information from many channels), LYSO also comes with a significant physical hurdle. It contains about 2.6% ^{176}Lu , a radioactive isotope that β -decays, producing a high intrinsic background that drowns out genuine signals from other sources in the same energy range. Our primary challenge is essentially to distinguish genuine physical signals from this background by developing suitable data acquisition and background suppression methods.

In Section 2 we describe our hardware setup and trigger system, whose general features are common to all measurements performed thereafter. Before dealing with the LYSO detector, our strategy was to start with a simpler reference system: a single-channel NaI scintillator coupled to a PMT. After calibrating and characterising the NaI PMT using known radioactive sources (described in Section 3), we exploit the back-to-back photon emission of a ^{22}Na source to establish a coincidence trigger between the NaI and LYSO scintillators. By doing so, the NaI acts as a robust background-suppression system, allowing for clean measurements from the more complicated LYSO system.

We then focused on the LYSO PMT alone (Section 4). We studied its response to variation in High Voltage (HV), performed energy calibration and characterised the spatial distribution of scintillation light across neighbouring channels. Then we investigated two strategies to suppress the intrinsic background without relying on the NaI coincidence: a topological filter based on charge sharing patterns between neighbouring channels and a timing cut based on Time-to-Digital Converter (TDC) information.

Finally, in Section 5 we discuss the limitation of the methods presented in this work and propose possible improvements for future measurements with the same setup.

2 Experimental setup

In this section we describe the experimental setup and hardware, and summarise the main operating parameters (e.g. HV, pedestal current, ...) used throughout the rest of this work.

2.1 Hardware

The experimental setup comprises three subsystems: radiation detectors, NIM electronics for signal processing, and a data acquisition system (DAQ). The manual for each module is listed in [A](#).

Detectors

- **Nal(Tl) scintillator:** a cylindrical inorganic scintillator (Nal(Tl)) coupled to a Photomultiplier Tube (PMT) (model 9266B [2]) of 51 mm diameter.
- **LYSO detector:** a $(Lu_{1-x}Y_x)_2SiO_5$ crystal array coupled to a flat-panel multianode PMT (model H8500 [1]) with a $49 \times 49 \text{ mm}^2$ active area (external size $52 \times 52 \text{ mm}^2$) and an 8×8 matrix of independent anode channels.

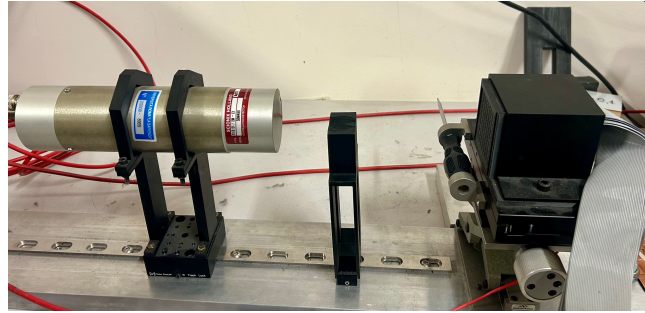


Figure 1: (Left) Nal(Tl) detector; (right) LYSO detector.

NIM modules

- **Discriminator (CAEN N417) [3]:** an 8-channel module that converts analogue pulses into standard NIM logic signals. It accepts both negative and positive input pulses (with absolute values from 1.5 mV to 250 mV), with manually adjustable voltage thresholds and output widths, set via the front-panel "THR" and "WDT" trimmers.
- **Dual Timer (CAEN 2255A) [4]:** a triggered pulse generator, used to define the DAQ integration gate by producing NIM pulses with adjustable widths ranging from 50 ns to 10 s.
- **3-Fold Logic Unit (CAEN N405) [5]:** a logic module used to perform coincidence operations between signals of different channels and gates.
- **Delay lines:** more specifically a 20-channel line providing 50 ns of delay, and two further 12-channel lines providing 100 ns and 50 ns of delay respectively.

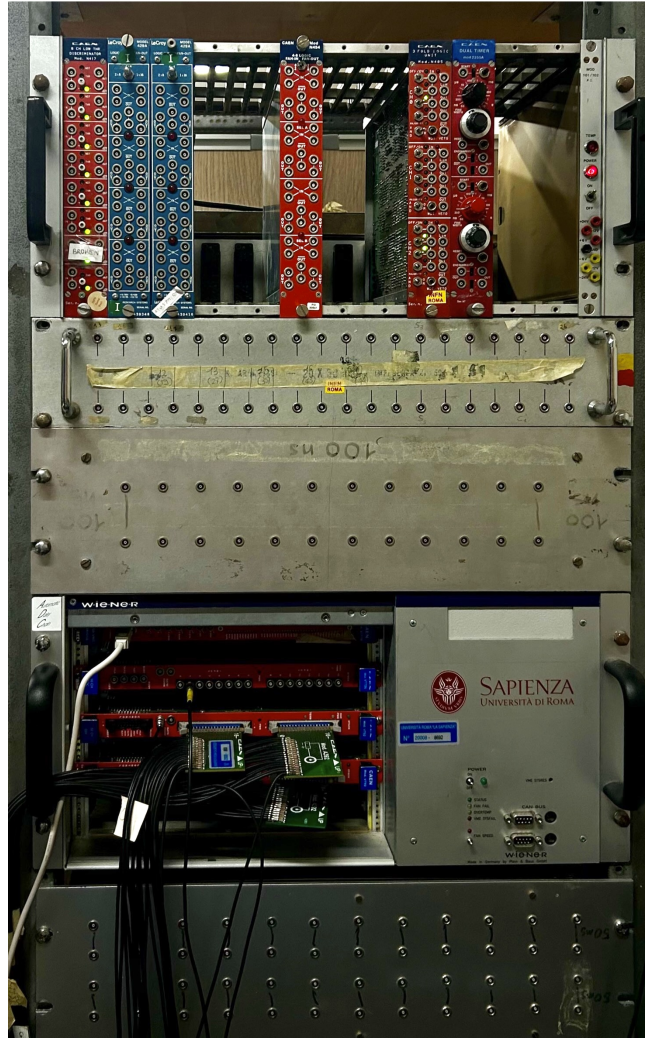


Figure 2: The crate houses NIM modules at the top, delay lines, and VME modules.

DAQ system

- **Charge-to-Digital Converter (QDC, CAEN V792)** [6] : a 32-channel VME module that integrates negative input currents. Each channel is routed through a Charge-to-Amplitude Conversion (QAC) stage; the QAC outputs are multiplexed and subsequently digitized by two fast 12-bit ADCs.
- **Time-to-Digital Converter (TDC, CAEN V775N)** [7] : a 16-channel VME module that, in COMMON START mode (used throughout this work), measures time intervals between a common signal (COM) and each input signal. Each time interval is first converted into a voltage level by a Time-to-Amplitude Converter (TAC); the TAC outputs are then multiplexed and digitised by two internal ADCs.
- **VME-USB Bridge (CAEN V1718)** [8]: reads out the QDC and TDC data over the VME bus and transfers it to the computer via USB.

Auxiliary instrumentation

- **Oscilloscope (Tektronix TDS 520D)** [10]: used for pulse-shape monitoring and timing verification before VME digitisation.
- **HV Power Supply (CAEN DT8033M)** [9] :powers the PMT and multi-anode PMT through two independent output channels, each set to its own operating range.

- **Function generator (GW InsteK GFG-8015G)** [11]: generates regular pulse signals with adjustable frequency (up to 2 MHz), used as a random trigger for the pedestal calibration.

Detection principle

Three radioactive sources were used for energy calibration and resolution characterisation of the detectors. Source names and ID codes are given in Table 1.

Source	ID Code
^{137}Cs	GN265
^{60}Co	GN292
^{22}Na	AH1805

Table 1: Radioactive sources used for calibration.

When a photon from one of these sources interacts with the NaI or LYSO crystals, the scintillation process produces light that the PMT converts and amplifies into an analogue electrical pulse, with amplitude proportional to the energy deposited in the crystal. This analogue signal is first sent to the NIM discriminator, which acts as a first filter against electronic noise, and produces a NIM logic pulse only for events above a set voltage threshold. The discriminator also copies the original analogue signal, which is delayed and sent to the QDC for charge integration. The discriminator’s logic output is then processed by a chain of NIM modules that implement the trigger/selection logic (described per configuration below). The Dual Timer generates an integration window (the *gate*) that defines the time interval during which the QDC and TDC modules operate. The entire front-end setup is powered by the HV supply. Communication with the computer is managed by the CAEN V1718 VME bridge. The two acquisition modules provide complementary information:

- the QDC outputs digital values corresponding to the integral of the detector current, plus a constant pedestal current (see Section 2.3 for details), within the *gate*;
- the TDC outputs digital values corresponding to the time difference between the *gate* signal and each discriminator signal.

For each triggered event, it is possible to record integrated charge for up to 32 channels, and time intervals for up to 16 (although not as many are used in practice). These data are stored in a *.root* file for data analysis.

2.2 Configuration settings

Key factors in the acquisition and trigger process include the discriminator thresholds, the data-acquisition gate width, and the power-supply settings.

- The discriminator voltage thresholds must be low enough to let the genuine signal through, but not so low that background noise dominates. As observed on the oscilloscope, a typical pulse from a ^{22}Na source has an amplitude of about 80 mV for the NaI scintillator operating at 550 V (baseline: 17 mV), and 400 mV for the LYSO detector operating at 980 V (baseline: 22 mV). The background noise (pedestal), measured with the chosen $I_{ped} = 100$, was of order 1 mV with fluctuations of order 0.02 mV (standard deviation). Hence, the thresholds were set to -25 mV for the NaI scintillator and -150 mV for the LYSO detector.
- The gate used for the QDC and TDC must be long enough to fully contain a single physical pulse, but not so long that it risks integrating a second, unrelated event (pile-up) or excess noise. A pulse from the NaI

PMT has a typical width of about 300 ns (FWHM), while the LYSO PMT produces significantly shorter pulses of order 50 ns (FWHM).

Since the gate is common for both detectors, its width was set by the slower NaI signal: it was fixed at 400 ns. Figure 3 shows the alignment of the NaI signal within the gate as observed during setup.

- The criteria used to determine the optimal HV value for each detector are detailed in Sections 3.1 and 4.4.

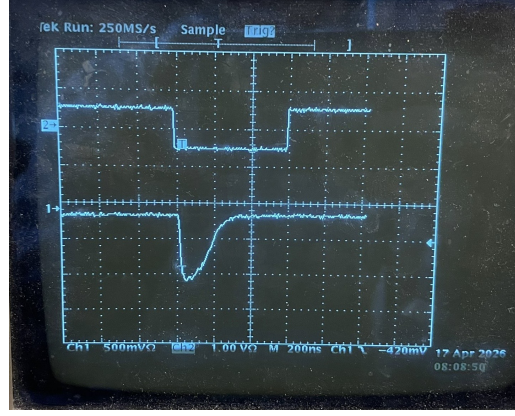


Figure 3: Oscilloscope screen: analogue signal within the gate, here set at 600 ns.

Relevant values are summarised in Table 2.

Parameter	Value
NaI discriminator threshold	−25 mV
LYSO discriminator threshold	−150 mV
NaI HV	550 V
LYSO HV	980 V
Gate width	400 ns

Table 2: Summary of operating parameters.

2.3 Pedestal calibration

As noted in Section 2.1, the QDC only integrates negative input currents. A genuine physical signal consists of a negative pulse superimposed on electronic noise fluctuating around zero. Without any compensation, the fraction of this noise that stays near or above zero current would be truncated by the Analog-to-Digital Converter (ADC), which cannot measure negative counts (*underflow*). To prevent this, the QDC adds a constant pedestal current (I_{ped}) to every channel’s input before integration, shifting the noise distribution to a baseline within the ADC’s dynamic range. I_{ped} is set once per board via a software register, but because of hardware variations between channels, the resulting offset is not identical across channels even for a fixed I_{ped} , and must be measured and subtracted individually for each channel (this analysis is performed below).

To determine the optimal value of I_{ped} , we studied the charge (in ADC counts) as a function of this parameter. The measurement was performed using a pulse generator connected to the start input of the Dual Timer, which periodically opened a 400 ns integration gate uncorrelated with physical detector events (a random trigger), allowing us to sample the baseline noise directly.

For QDC channel 0, the resulting curve and the pedestal distribution at the chosen operating point are shown in Figure 4.

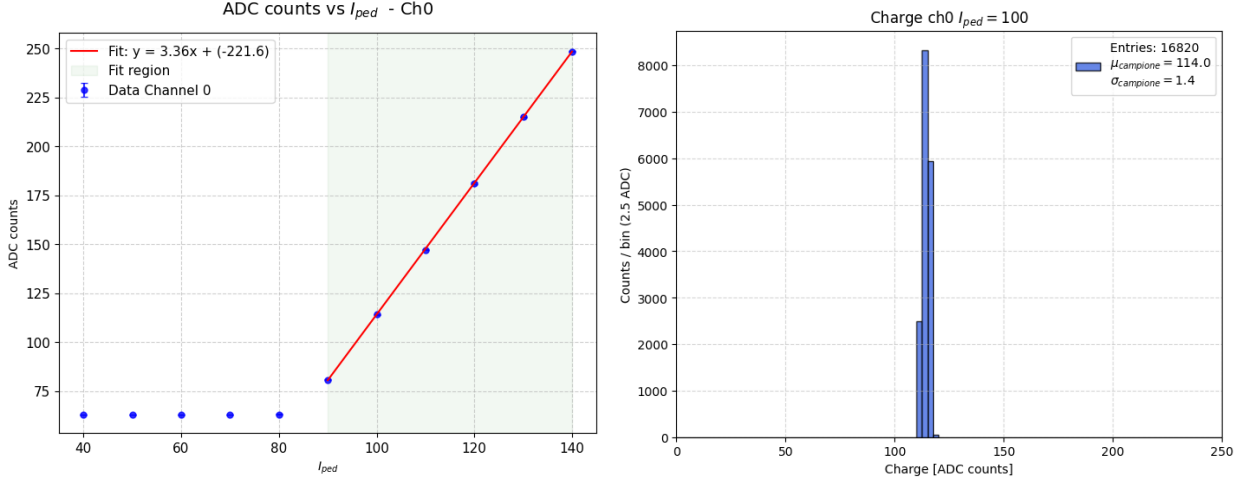


Figure 4: (Left) ADC counts as a function of I_{ped} for channel 0, with linear fit ($y = mx + q$) for $I_{ped} \geq 90$. (Right) Integrated charge at the selected operating point ($I_{ped} = 100$) for channel 0.

m	q [ADC]
3.358 ± 0.002	-221.6 ± 0.3

Table 3: Parameters of the linear fit for $I_{ped} > 90$.

Below $I_{ped} \approx 80$, the response is completely flat (Figure 4, left): this is the underflow region described above, where noise is truncated by the ADC. Increasing I_{ped} beyond ≈ 90 moves the QDC into a linear regime.

We therefore selected $I_{ped} = 100$ as the operating point, which lies safely within the linear region and corresponds to a nominal physical current of $18.04 \mu\text{A}$ ¹. At this operating point, the noise can be thought of as a Gaussian distribution (Figure 4, right) with mean $\mu = (114.25 \pm 0.01)$ and $\sigma = (1.40 \pm 0.01)$ ADC counts.

For the following data analysis, the uncertainty on each individual charge measurement will have to account for this intrinsic pedestal resolution. This calibration procedure was repeated for every channel of the QDC. Each channel's pedestal cannot be assumed to be stable over time and in changing environmental conditions (mainly temperature). To account for this, a pedestal run was taken shortly before each new data acquisition (see Appendix A).

¹Value taken from the V792 module manual [6].

3 Characterisation of NaI detector

Due to its well-defined response and low background noise (compared to LYSO), the NaI detector serves as a simple reference system for the more complicated LYSO case. In this section, we describe the experimental procedures and techniques used to characterise the NaI. In particular, we analyse the PMT's response to variation of HV, and then we perform energy calibration. In the process, the energy spectra of the radioactive sources listed in Tab 1 are acquired.

3.1 Gain-HV dependence

According to the manufacturer's specifications ², the operating voltage range for the NaI PMT extends from 500 V to 800 V. To identify the optimal HV value, a HV scan was performed using a ^{137}Cs radioactive source, chosen for its high activity and well-defined photopeak at around 661.7 keV. For each voltage value, the source's energy spectrum was acquired in a 500 s run, with the number of collected events increasing from $\sim 6.9 \times 10^4$ at 500 V to $\sim 8.6 \times 10^5$ at 600 V. A background spectrum was also acquired for each HV value, over the same time interval of 500 s, with the source removed and all other conditions unchanged. Since the two runs had equal duration, the background spectrum was subtracted bin-by-bin from the source spectrum without any rescaling. Photopeaks were then extracted from the background-subtracted spectra via Gaussian fit, restricted to a narrow window around the photopeak. Peak positions are shown as a function of the HV in Figure 5. A fit with the exponential function $y = Ae^{bx}$ returns the parameters in Table 4.

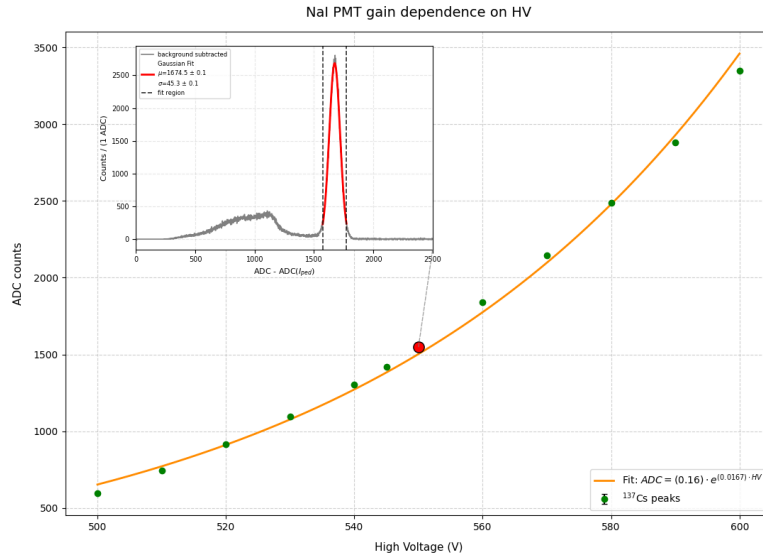


Figure 5: Position of the ^{137}Cs peak as a function of applied HV. The peaks are extracted from background-subtracted spectra via Gaussian fit, an example of which is displayed in the inset for $\text{HV} = 550\text{V}$ (the fit region was chosen to exclude the Compton edge at lower energies). The points are then fitted with an exponential function $y = Ae^{bx}$. Uncertainties on the peak positions correspond to the μ uncertainty returned by the Gaussian fit.

A [ADC]	b [V^{-1}]
0.16 ± 0.03	0.0167 ± 0.0004

Table 4: Parameters extracted from the exponential fit.

²Details in the PMT manual (mod 9266B) [2].

Since the ADC output for a fixed-energy peak is proportional to the PMT gain, this increasing trend essentially represents the gain-HV dependence. Within the explored voltage range, it can be locally approximated by an exponential function $y = Ae^{bx}$ ³. Figure 6 illustrates the effect of increasing supply voltage on the energy spectrum of ^{137}Cs . At 590 V, the spectrum is significantly stretched and shifted towards higher ADC channels compared to the spectrum at 540 V. This shift expands the dynamic range for low-energy signals but risks saturating the QDC for higher-energy events.

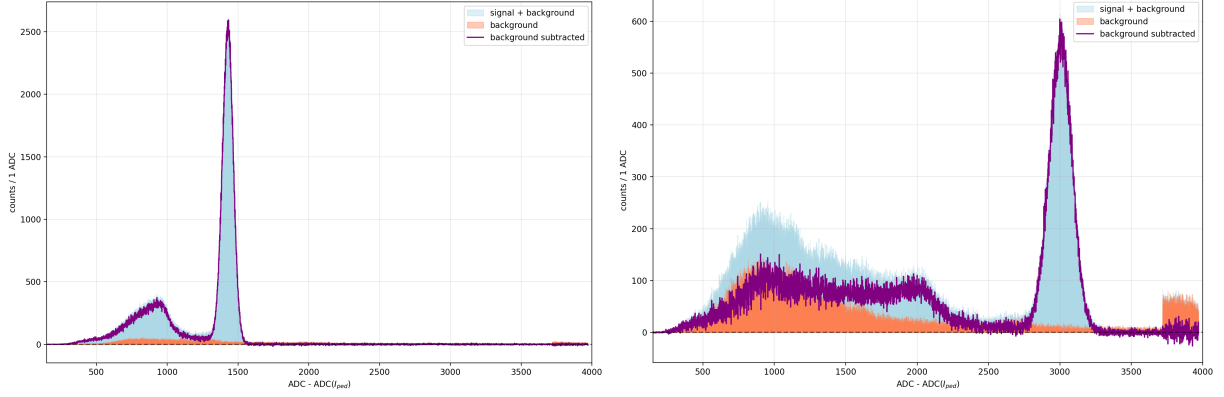


Figure 6: ^{137}Cs spectra measured on NaI PMT with HV respectively of 540 V (left) and 590 V (right). Notice the shift of the photopeak towards higher ADC channels as the HV increases. In both plots, notice the pile-up of events at the upper end of the ADC range, consistent with high-energy events saturating the QDC dynamic range (likely cosmic rays).

A reasonable criterion for choosing the operating HV is to use the highest value that does not saturate the sources of interest. Hence, the HV for the NaI PMT was set to 550 V, which provides enough gain to clearly resolve the ^{60}Co peaks (the highest among our selection) without saturating the QDC.

3.2 ADC to energy calibration

After setting the HV, the NaI detector was calibrated using three radioactive sources with well-known emission lines: ^{137}Cs , ^{22}Na and ^{60}Co . For each source the background-subtracted energy spectrum is acquired just as above, for ^{137}Cs , ^{60}Co and ^{22}Na respectively (Figures 7, 8 and 9).

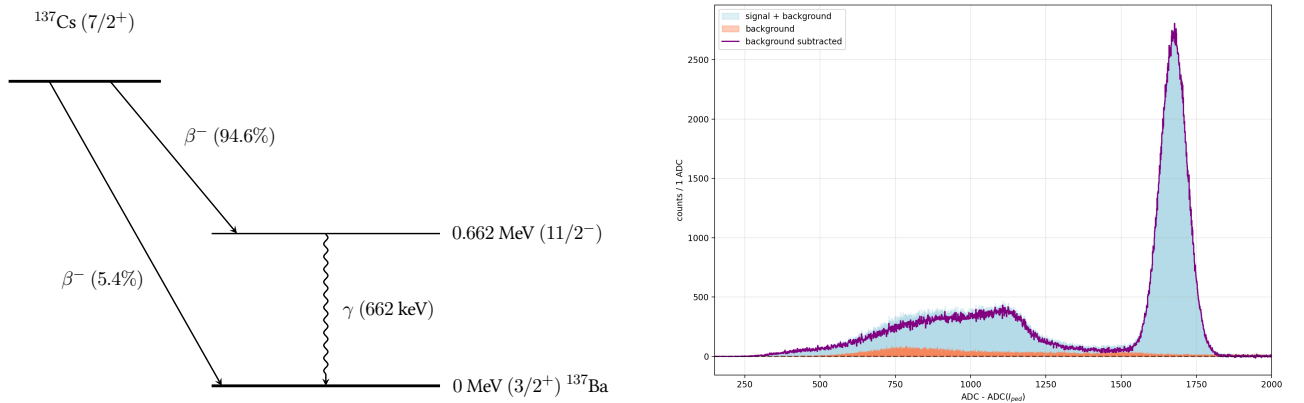


Figure 7: (Left) decay scheme of the ^{137}Cs source showing the main β^- decay branch followed by the 662 keV gamma emission. (Right) the experimental ^{137}Cs energy spectrum (background subtracted). The run with the source returned 582444 events in a 500 s runtime.

³PMT gain theoretically follows a power law, $G \propto V^k$, where k depends on the number of dynode stages. Over the limited voltage range explored here (500 V to 600 V), an exponential function is a good local approximation.

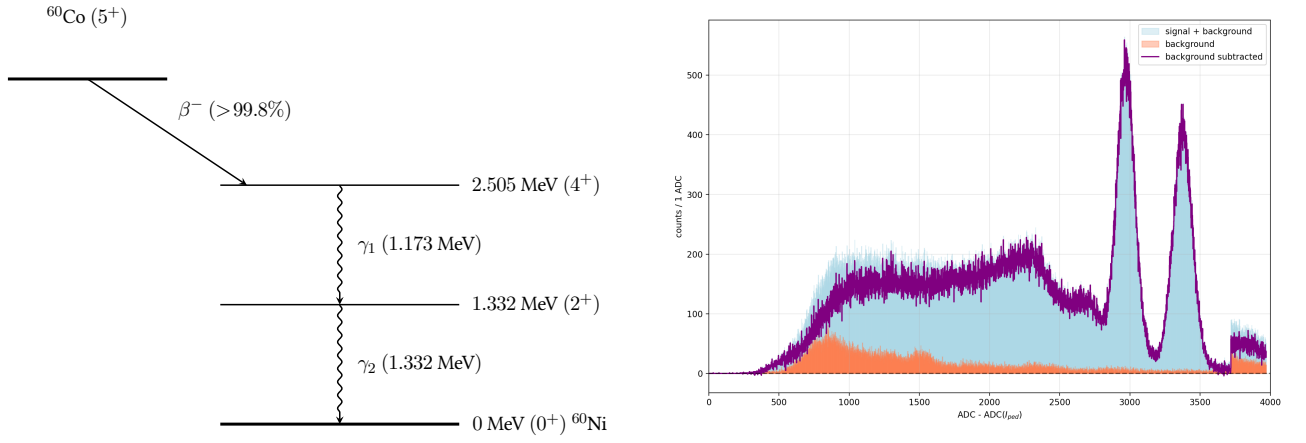


Figure 8: (Left) the decay scheme of the ^{60}Co source showing the main β^- decay branch followed by the characteristic two-photon gamma cascade. (Right) the experimental ^{60}Co energy spectrum (background subtracted). The run with the source returned 549851 events in a 500 s runtime.

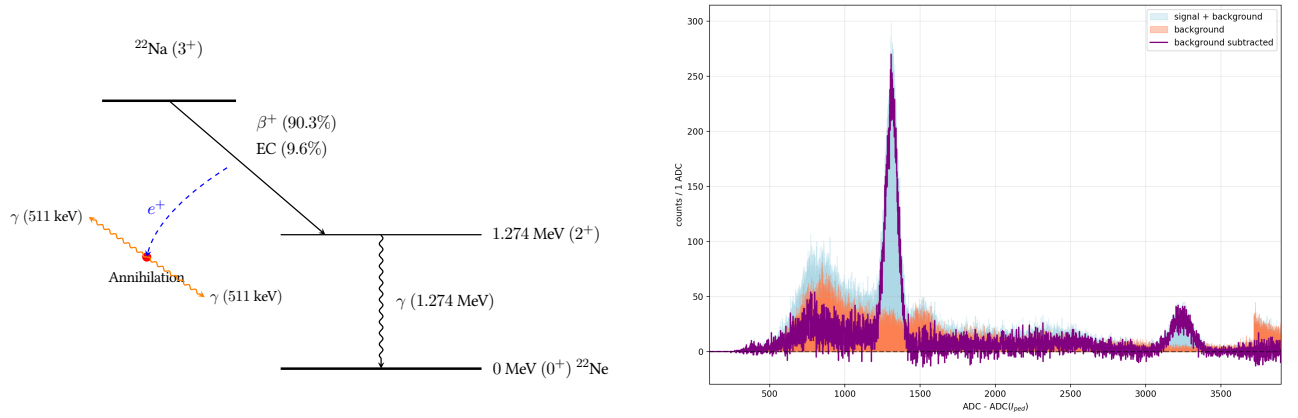


Figure 9: (Left) the decay scheme of the ^{22}Na source showing the positron annihilation into two back-to-back 511 keV photons. (Right) the experimental ^{22}Na energy spectrum (background subtracted). The run with the source returned 106971 events in a 500 s runtime.

The position of each photopeak was then extracted via Gaussian fit. The chosen fit ranges are highlighted in Figure 10 and Figure 11.

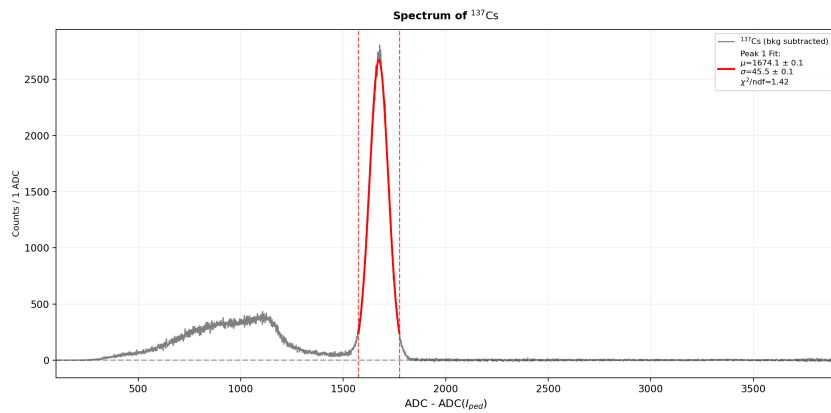


Figure 10: Background-subtracted spectrum of ^{137}Cs source. Grey lines show background-subtracted data, while the red curve represents the Gaussian fit of the photopeak, performed within the highlighted region.

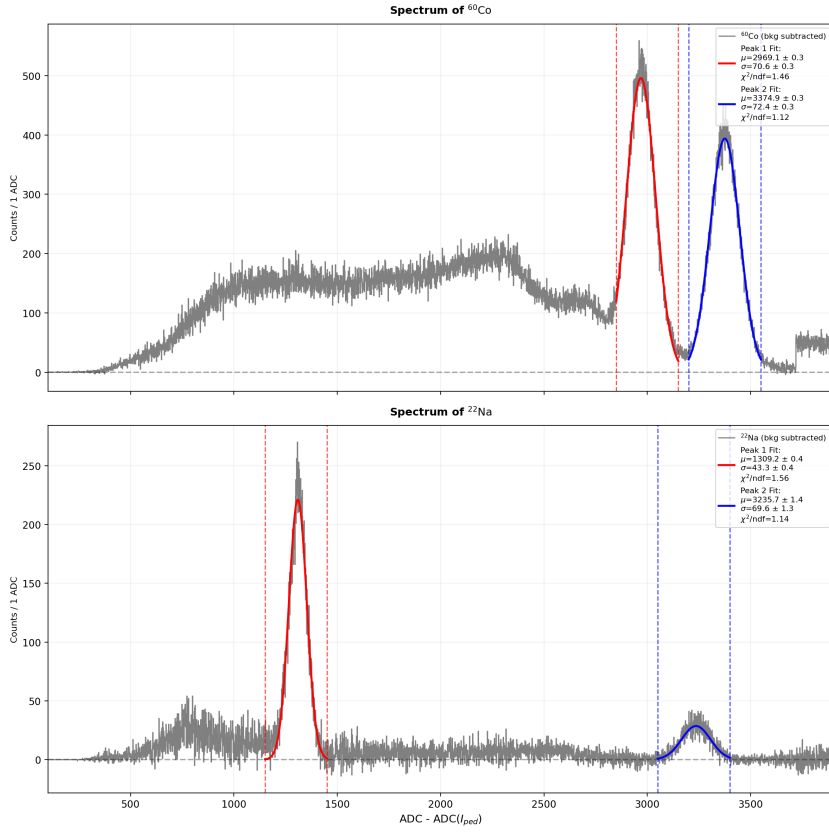


Figure 11: Background-subtracted spectra of ^{60}Co and ^{22}Na sources. Grey lines show background-subtracted data, while red and blue curves represent the Gaussian fits of the photopeaks, performed within the highlighted regions.

The fit results are summarised in Table 5.

Source	Peak [keV]	Fit Range [ADC]	μ [ADC]	σ [ADC]	χ^2/ndf
^{137}Cs	662	1574 – 1774	1674.1 ± 0.1	45.5 ± 0.1	$\frac{280.16}{197} = 1.42$
^{60}Co	1173	2850 – 3150	2969.1 ± 0.3	70.6 ± 0.3	$\frac{432.71}{297} = 1.46$
	1332	3200 – 3550	3374.9 ± 0.3	72.4 ± 0.3	$\frac{388.28}{347} = 1.12$
^{22}Na	511	1150 – 1450	1309.2 ± 0.4	43.3 ± 0.4	$\frac{463.20}{297} = 1.56$
	1275	3000 – 3450	3236 ± 1	70 ± 1	$\frac{395.25}{347} = 1.14$

Table 5: Fit parameters, χ^2/ndf , and fit ranges for the Gaussian photopeaks of ^{137}Cs , ^{60}Co , and ^{22}Na . The reduced χ^2 are all below 2, suggesting that the single Gaussian model provides a correct description for each peak within the selected range.

Fitting the measured ADC counts against the known photopeak energies yields the calibration plot shown in Figure 12, with fit parameters given in Table 6.

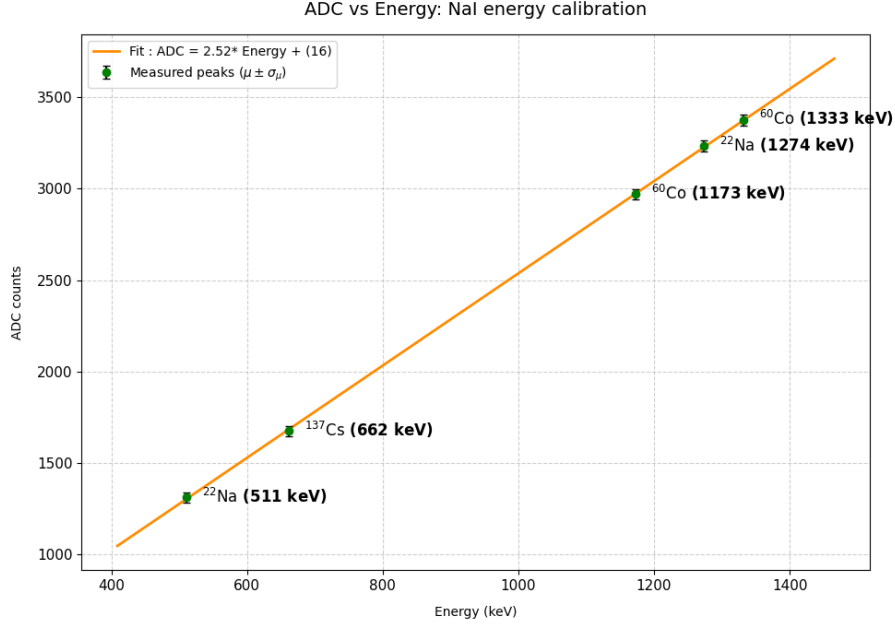


Figure 12: Energy calibration for NaI detector. The linear fit function used is $y = mx + q$. The uncertainties on the measured points correspond to the uncertainties on the mean returned by the Gaussian fits of the photopeaks.

m [ADC/keV]	q [ADC]
2.52 ± 0.01	16 ± 9

Table 6: Parameters of the NaI PMT linear calibration fit.

These parameters allow us to convert ADC counts from any QDC measurement into energy (keV), and vice versa.

Isotope	Energy [keV]	Peak Position [ADC]
^{22}Na	511.0	1312 ± 30
^{137}Cs	661.7	1674 ± 30
^{60}Co	1173.2	2969 ± 30
^{22}Na	1274.0	3233 ± 30
^{60}Co	1332.5	3375 ± 30

Table 7: Peak positions of radioactive sources. Here peak position uncertainty is the quadrature sum of the Gaussian fit error ($\sigma_{G_{fit}}$), pedestal noise (σ_{ped}), and HV-induced fluctuations (σ_{ADC}). The HV contribution is the dominant source of uncertainty. Since σ_{ADC} (Eq. 3.2) depends only on the fixed operating HV and not on the individual peak, it is identical for all rows, which is why the total uncertainty is the same across all five measurements.

The total uncertainty associated with each peak is given by:

$$\sigma_{peak} = \sqrt{\sigma_{G_{fit}}^2 + \sigma_{ped}^2 + \sigma_{ADC}^2}$$

where the three terms summed in quadrature are:

- $\sigma_{G_{fit}}$, the uncertainty on the mean value returned by the Gaussian fit performed on the photopeaks;

- σ_{ped} , the pedestal resolution as determined in Section 2.3, which accounts for the baseline electronic noise of the QDC;
- σ_{ADC} , the uncertainty related to the HV power supply.

σ_{ADC} is obtained by propagating the uncertainty on HV through $ADC = Ae^{bHV}$ (see Section 3.1), and is therefore strongly dependent on the chosen HV:

$$\sigma_{ADC} = A b e^{bHV} \sigma_{HV}$$

Here σ_{HV} accounts for the uncertainty of the power supply ($\pm 0.02\%$ of the set HV value + 1 V). In our setup, the dominant contribution to peak uncertainty comes from σ_{ADC} , while σ_{ped} and $\sigma_{G_{fit}}$ are small corrections.

Following the energy calibration, the same background-subtracted spectra can now be expressed in terms of physical energy rather than ADC counts, as shown in Figure 13.

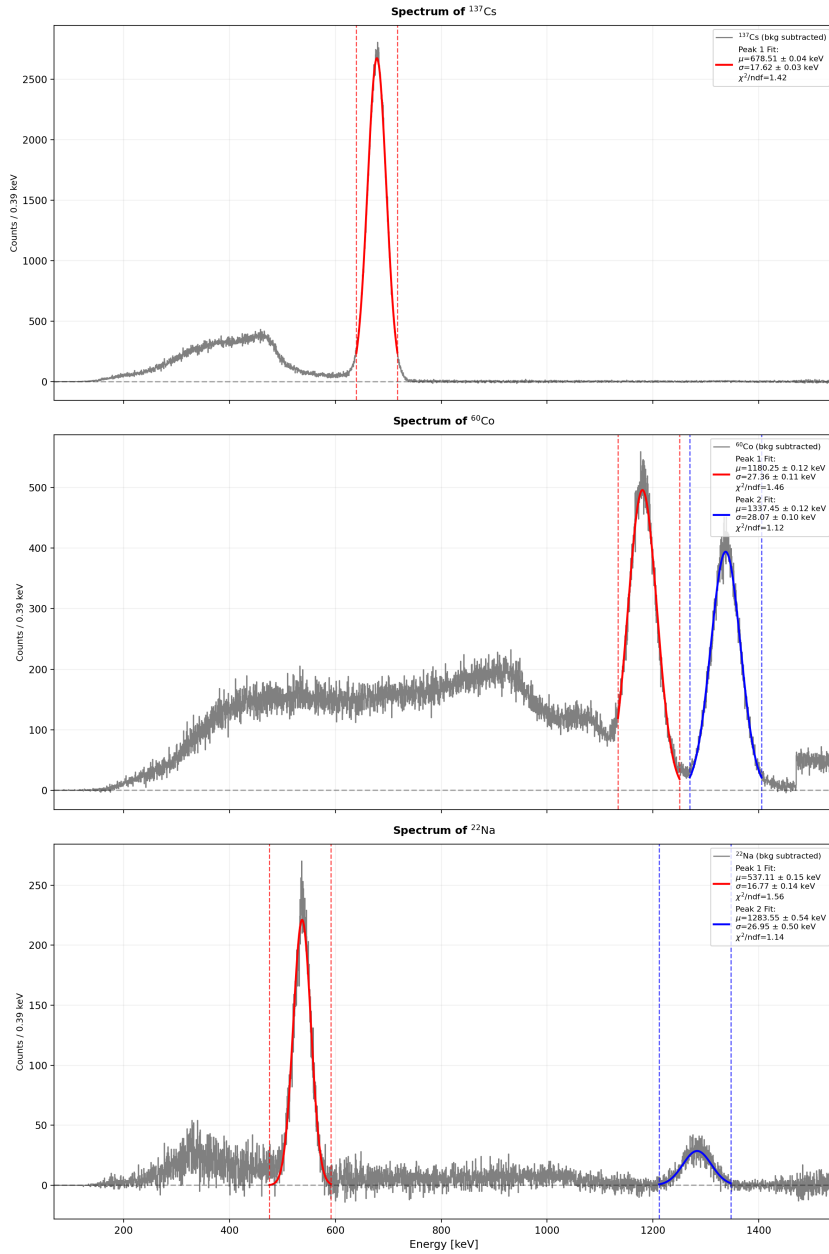


Figure 13: Background-subtracted spectra of ^{137}Cs , ^{60}Co and ^{22}Na sources. The x-axis represents physical energy [keV].

To characterise the energy resolution of the NaI detector, the peak widths (converted from ADC counts to keV using the slope of the energy calibration) were analysed as a function of the corresponding gamma-ray energies. For each photopeak, the parameter σ and its uncertainty, extracted from the Gaussian fits of the peaks (Figure 11), were used to compute the relative resolution of the detector, σ/E , where E is the tabulated energy of the corresponding gamma-ray line. Since the uncertainty on E is negligible compared to the one on σ , the uncertainty on σ/E reduces to the uncertainty of the σ parameter returned by the fit.

Figure 14 shows σ/E as a function of E .

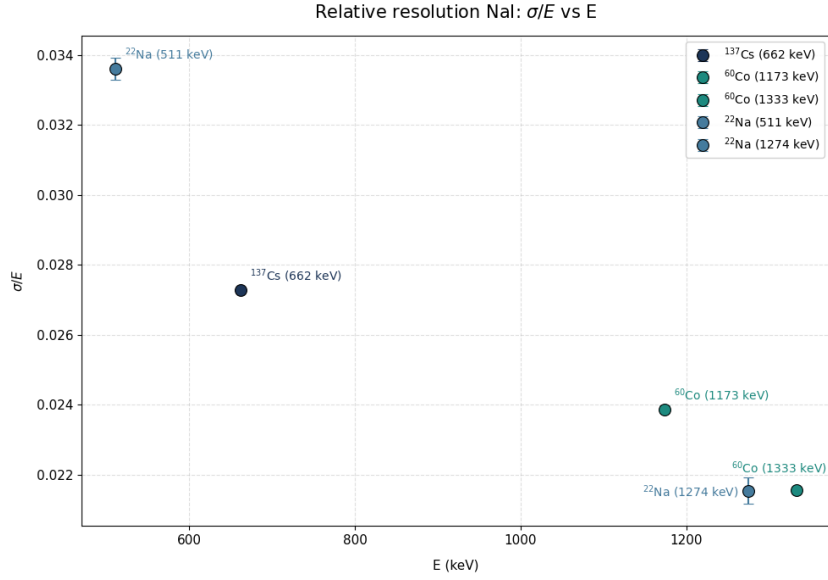


Figure 14: σ/E vs E for the NaI detector.

The data follows the expected qualitative behaviour for an NaI scintillator: the relative resolution improves (i.e. σ/E decreases) with increasing energy, consistent with a resolution dominated by photoelectron statistics, which scales approximately as $\frac{1}{\sqrt{E}}$.

4 Characterisation of LYSO detector

As noted in Sec. 1, the main physical limitation of a LYSO scintillator is the fact that it contains about 2.6% of the ^{176}Lu isotope, whose radioactive decay produces a significant level of intrinsic background.

The primary challenge in using such a scintillator is therefore to distinguish genuine physical signals from this intrinsic background by developing an appropriate background suppression strategy.

In addition to this physical limitation, the particular LYSO detector used in this experiment presents a technical challenge arising from its highly segmented readout. The 23×23 pixel scintillator is coupled to a 64-channel multi-anode PMT, requiring the signals from each channel to be mapped, calibrated and processed independently. Furthermore, the scintillation light produced by a single event is typically shared among several neighbouring PMT channels (as opposed to a single channel in the NaI case), making event reconstruction non-trivial.

In this section, we begin by describing the mapping of PMT channels onto QDC/TDC inputs, as well as the trigger logic and event reconstruction procedure. We then discuss the dependence of PMT gain on applied HV and the calibration relating measured ADC counts to deposited energy. In the process, several background suppression techniques are suggested and used, to varying degrees of effectiveness:

1. coincidence measurements using back-to-back 511 keV photons from a ^{22}Na source and an external NaI scintillator (AND logic);
2. filtering based on charge-sharing topology;
3. discrimination based on timing information provided by the .

4.1 Pin mapping, trigger logic, runs

Figure 15 shows the front face of the 8×8 multi-anode PMT (coupled to the LYSO scintillator) in the two configurations used in the experiment. Each anode (hereafter referred to as a *pin*) has an active area of $5.8 \times 5.8 \text{ mm}^2$ ⁴.

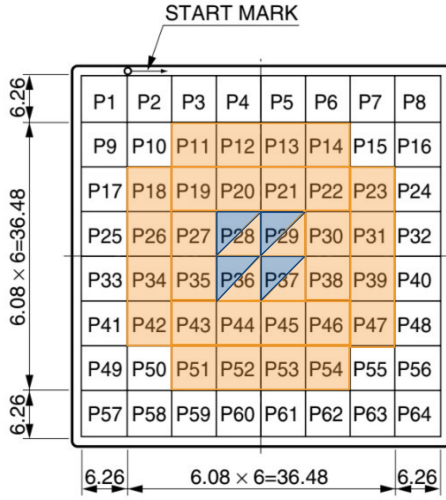
In an ideal setup, each of the 64 pins would be connected to both a QDC and a TDC, and would also contribute to the trigger logic through a global OR. Such a configuration would provide complete charge and timing information for every scintillation event. However, this would require 64 inputs (only 16 were available), 64 discriminator channels (7 available), and a correspondingly large logic unit and number of cables. While technically feasible, such a setup was beyond the scope of this experiment and unnecessary for the goal of characterising the detector pursued here.

The first objective was to determine the spatial distribution of scintillation light on PMT pins for single events; in particular, how many neighbouring pins must be integrated over in order to recover the entire charge produced by a single interaction.

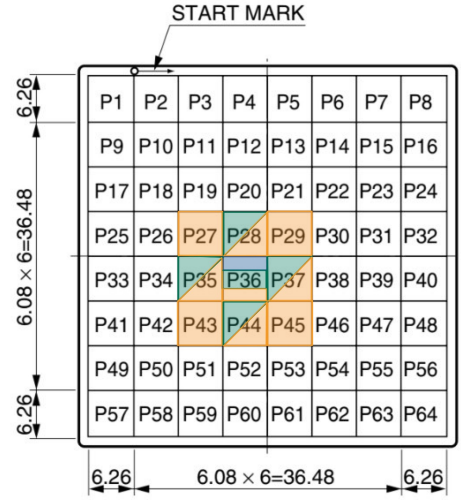
To this end, and due to the hardware constraints described above (most importantly the limited number of discriminator channels available on the CAEN N417), the detector was initially set up in *Configuration A* (see Fig. 15), with 32 pins connected to the QDC.

In a series of test runs, radioactive sources were positioned at the centre of the LYSO matrix (to maximise solid angle) and the trigger was generated using the OR of four central PMT channels (pins 28, 29, 36 and 37, highlighted in blue in Fig. 15). Whenever one of these four channels measured a pulse exceeding a predefined threshold (set to -250 mV when $HV = 1100 \text{ V}$, and to -150 mV when $HV = 980 \text{ V}$), the discriminator generated a trigger signal that opened the QDC acquisition gate. During this gate, the charge recorded by every connected PMT channel was integrated and stored.

⁴Details provided in the H8500 multi-anode PMT manual [1].



(a) *Configuration A*: the trigger is generated by the OR of four central pins (chosen to maximise geometric acceptance); no TDC information is saved.



(b) *Configuration B*: the trigger is generated by pin 36 alone; TDC information is saved for four additional pins (excluding pin 36 itself).

Figure 15: LYSO pin connections in the two configurations used. Pins highlighted in yellow are read out via QDC; pins in green are read out via TDC; pins in blue are used for the trigger logic.

Comparing, for each event, the charge measured by the *central pin* (defined as the pin receiving the largest share of the event charge), its eight nearest neighbours, and the remaining pins, it became evident that the charge from a typical scintillation event is almost entirely contained within the central pin and a few of its nearest neighbours. We therefore established that, for every event, the charge integrated in a 3×3 pin cluster centred around the *central pin* is a reasonable estimator for the total charge of the event itself.

An example event is shown in Figure 16. Here, pin 37 is identified as the central pin, and the reconstructed event charge is obtained by summing the charges from pins 28, 29, 30, 36, 37, 38, 44, 45 and 46.

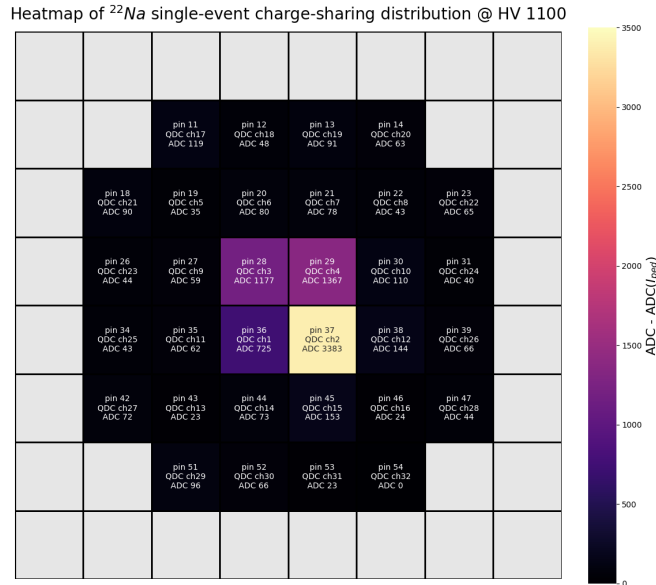


Figure 16: Example event display showing the charge distribution across LYSO pins for a single scintillation event (HV = 1100 V, used here to increase the event rate for illustration purposes). Pin 37 is identified as the central pin. The reconstructed cluster charge is the sum of the 3×3 grid of pins centred on pin 37.

It should be noted that the charge-sharing topology is not unique. Two further special cases are illustrated in Figure 17: (1) an event in which the central pin falls on the outer border of the instrumented region, so that part of the charge is (presumably) deposited in pins that are not read out; and (2) an event in which multiple interactions appear to occur simultaneously within the same gate. Events of type (1) are discarded *a posteriori*, since the cluster charge cannot be reliably reconstructed when boundary pins are missing.

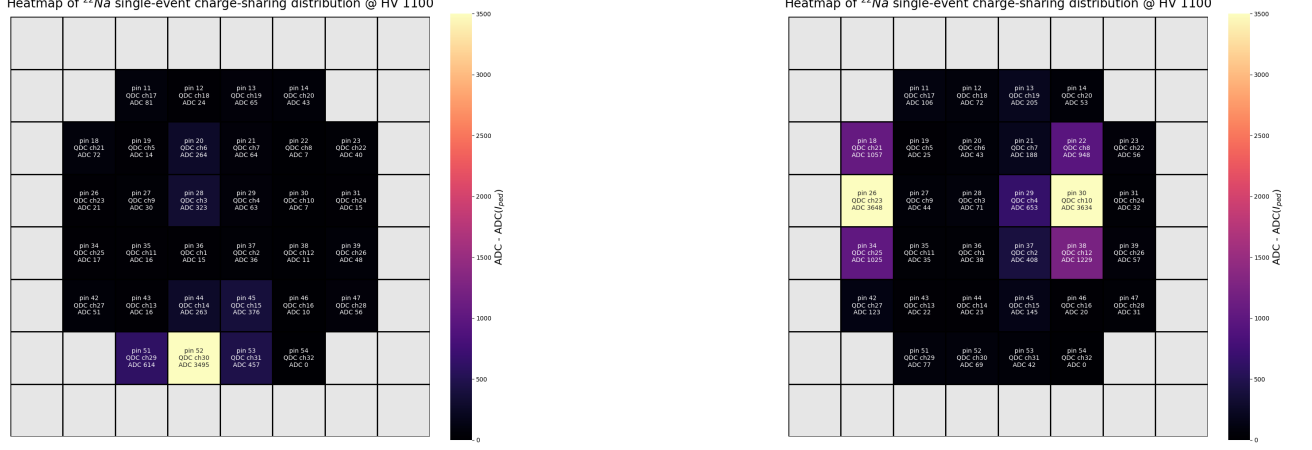


Figure 17: Further examples of event topologies (HV=1100 V). Left: an event whose central pin lies on the boundary of the instrumented region; such events are discarded. Right: an event consistent with two simultaneous interactions within the same acquisition gate.

Once timing measurements were introduced, the limited number of available channels became the dominant hardware constraint. This motivated the adoption of the alternative readout geometry shown in Fig. 15b (*Configuration B*). In this configuration, a single PMT channel (pin 36) provides the trigger. Both QDC and information are recorded for the five channels forming a cross centred on pin 36 (pins 28, 35, 36, 37 and 44), while the four corner channels of the 3×3 cluster (pins 27, 29, 43 and 45) are connected to the QDC only. In this way, the complete 3×3 cluster charge can still be reconstructed, while leaving sufficient channels available for the timing measurements discussed in the following sections.

In *Configuration B*, events for which the central pin does not coincide with pin 36 are discarded, as part of the charge would be deposited in pins outside the instrumented area.

4.2 ADC to energy calibration

The conversion between integrated charge (in ADC counts) and deposited energy was once again determined using the available radioactive sources: ^{22}Na , ^{137}Cs and ^{60}Co . In particular, measurements were taken in four runs (summarised in Tab. 8). In each run, the trigger was acquired from LYSO pin 36 alone in *Configuration B* (see Fig. 15).

Aside from technical complications arising from the highly segmented LYSO matrix, the calibration procedure is conceptually identical to that used for the NaI detector (Sec. 3.2): the photopeaks of the calibration sources were identified and fitted from their background-subtracted spectra, and the resulting ADC values were correlated with the known gamma-ray energies.

In practice, the procedure is complicated by the intrinsic radioactivity of LYSO: longer acquisition times are required in order to obtain spectra with a sufficiently high signal-to-background ratio. The background-subtracted spectra for the three calibration sources are shown in Figs. 18, 19 and 20, together with the background spectrum rescaled by a factor $\frac{\Delta t_{\text{signal}}}{\Delta t_{\text{bckg}}}$.

Run label	Source	Runtime	Number of events
bckg	none	9 hours	2428413
Na-single	^{22}Na	9 hours	2464604
Cs-single	^{137}Cs	3 hours	844654
Co-single	^{60}Co	6 hours	1746123

Table 8: Overview of runs. In each run, the trigger was acquired from LYSO pin 36 alone in *Configuration B* (see Fig. 15). A longer runtime was chosen for ^{22}Na , as the background suppression methods (see below) were mainly tested on this source.

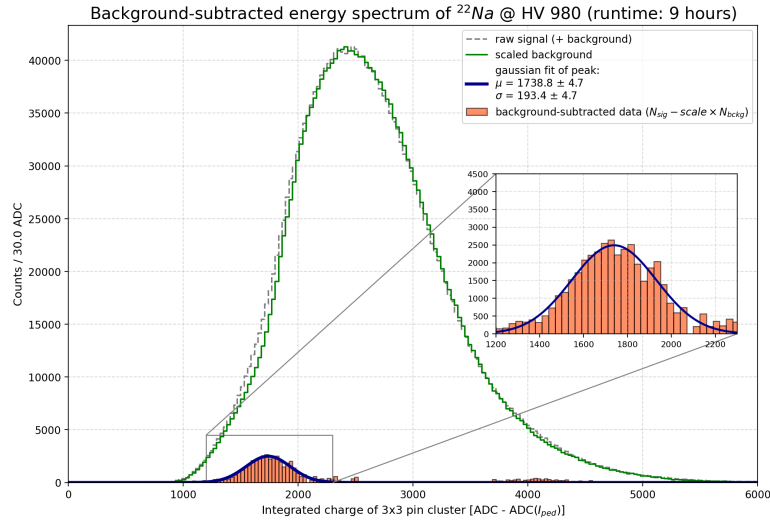


Figure 18: Background-subtracted spectrum acquired with the ^{22}Na source. Here signal (grey line) was acquired in a 9 hour run, and background (green line) in a 9 hour run. The second peak at 1275 keV is not resolved due to low signal-to-background ratio.

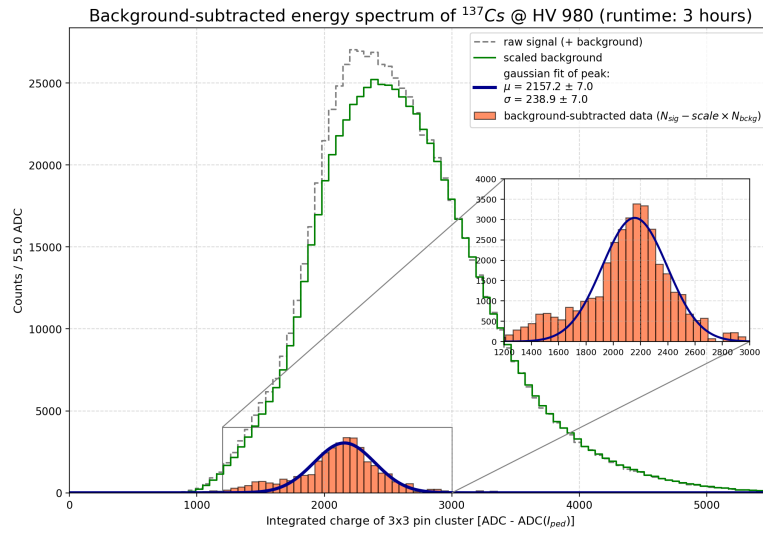


Figure 19: Background-subtracted spectrum acquired with the ^{137}Cs source. Here signal (grey line) was acquired in a 3 hour run, and background (green line) in a 9 hour run. In the plot, the background data was therefore rescaled by a factor $1/3$.

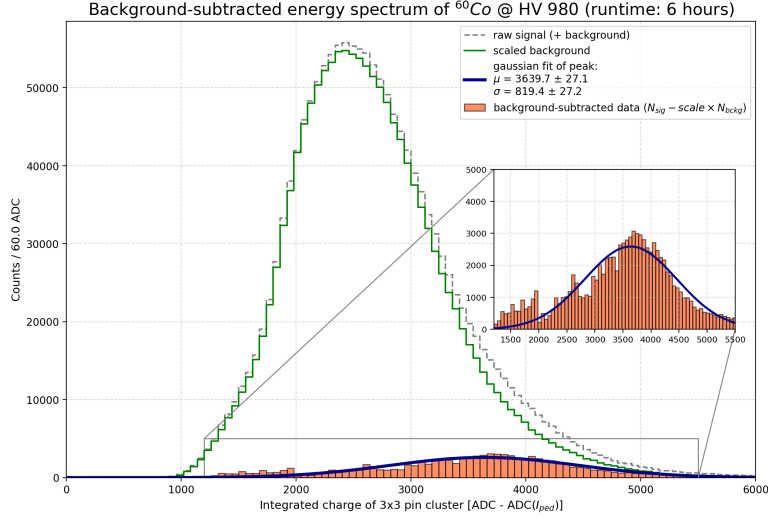


Figure 20: Background-subtracted spectrum acquired with the ^{60}Co source. Here signal (grey line) was acquired in a 6 hour run, and background (green line) in a 9 hour run. In the plot, the background data was therefore rescaled by a factor $2/3$. The pair of peaks at 1.173 MeV and 1.332 MeV are not resolved.

An important difference from the NaI case is observed in the ^{60}Co spectrum (Fig. 20). Whereas the NaI detector clearly resolves the two photopeaks at 1173 keV and 1332 keV, the energy resolution of the LYSO detector is insufficient to separate them, and a single broadened peak is observed instead. For calibration purposes, the centroid of this merged peak was associated with the mean of the two gamma-ray energies:

$$E_{\text{av}} = \frac{1}{2}(1173.2 + 1332.5) \text{ keV} = 1252.9 \text{ keV}.$$

The resulting ADC-to-energy calibration is shown in Fig. 21.

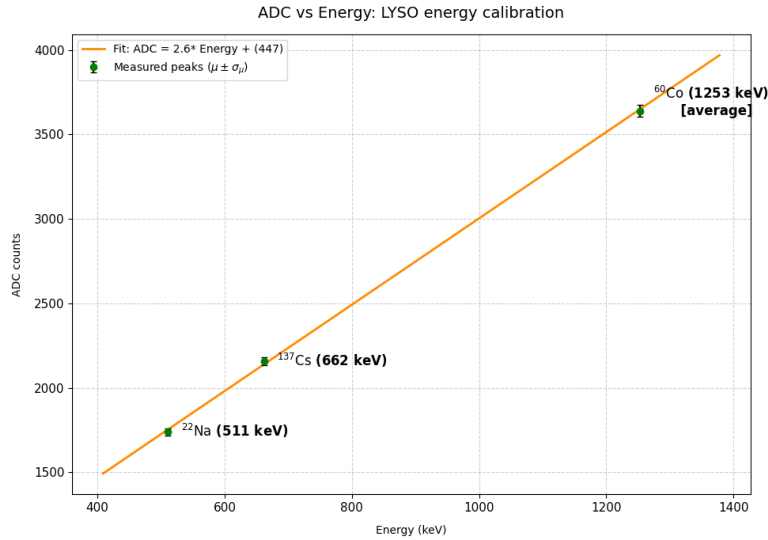


Figure 21: ADC-to-energy calibration for the LYSO detector, obtained from the fitted photopeaks of the three radioactive sources. The uncertainties on the measured points correspond to the uncertainty on the mean returned by the Gaussian fits of the photopeaks.

A linear fit $ADC = m(E) + q$ gives the parameters listed in Table 9. The calibration precision is primarily limited by the small number of calibration points and the comparatively large uncertainty associated with the unresolved ^{60}Co peak.

m [ADC/keV]	q [ADC]
2.6 ± 0.1	447 ± 45

Table 9: Parameters extracted from the linear calibration fit for the LYSO detector.

Isotope	Energy [keV]	Peak position [ADC]
^{22}Na	511.0	1739 ± 20
^{137}Cs	661.7	2157 ± 20
^{60}Co	1252.9*	3640 ± 30

Table 10: Peak positions of the radioactive sources used for the LYSO energy calibration, extracted from Gaussian fits. Uncertainties are computed with the Formula 3.2. *The ^{60}Co entry corresponds to the average of the two unresolved gamma-ray energies (1173.2 and 1332.5 keV).

The detector’s energy resolution was similarly analysed by plotting the relative resolution σ/E as a function of the gamma-ray energy (Figure 22).

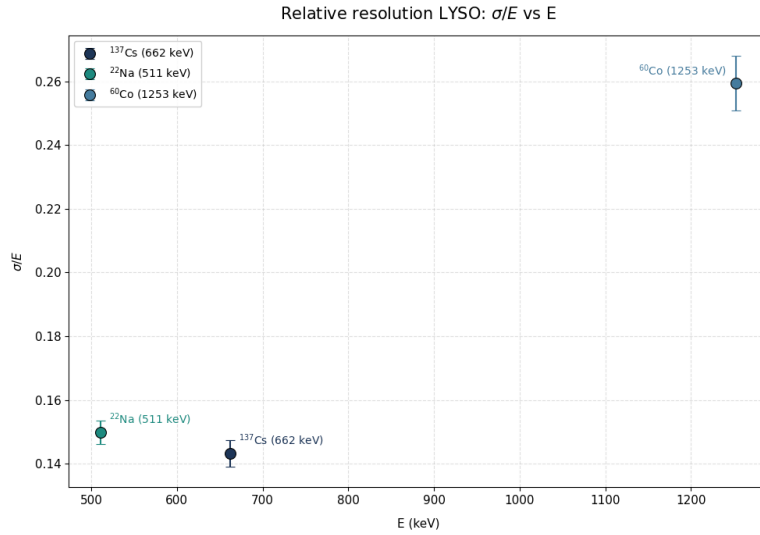


Figure 22: σ/E vs E for the LYSO detector. The uncertainties of σ parameters are extracted from Gaussian fit as the NaI case.

A striking difference is observed compared to the NaI case. While the NaI detector shows an improvement in resolution with energy (consistent with the $1/\sqrt{E}$ scaling). The LYSO data point for ^{60}Co at 1252.9 keV exhibits a significant increase in σ/E , deviating from the expected trend.

This should not be interpreted as a genuine worsening of the LYSO intrinsic resolution at high energies. Rather, it is an artefact of the merged Gaussian: since the two ^{60}Co gamma lines at 1173 and 1332 keV cannot be resolved by the LYSO detector, the fitted Gaussian σ reflects both the intrinsic resolution and the energy separation between the two lines. The extracted σ/E therefore overestimates the true detector resolution at this energy. More broadly, the LYSO energy resolution is worse than that of NaI across the entire measured range.

4.3 Background suppression via back-to-back coincidence

In the previous section, the most basic method of background suppression was used to resolve the source's energy peaks. Namely, a separate run was taken in the absence of sources, and the resulting spectrum was simply subtracted from the signal spectrum (up to rescaling due to differences in runtime). In this and the following sections, some alternative background-suppression methods are presented and discussed.

For measurements performed with the ^{22}Na source⁵, the decay kinematics provide a simple but powerful background suppression method. As seen in the decay scheme in Fig. 7, the positron emitted in the decay rapidly annihilates with an electron, producing two 511 keV photons emitted back-to-back. This can be exploited by placing the NaI detector opposite the LYSO detector and requiring a coincidence between the two. The trigger logic adopted is therefore:

$$(\text{NaIPMT}) \wedge (\text{LYSO pin 36}),$$

so that an event is recorded only when both detectors register a signal within the coincidence time window. Since the intrinsic ^{176}Lu activity of the LYSO crystal is uncorrelated with the NaI detector, this hardware coincidence rejects the vast majority of background events, making no further background subtraction necessary to resolve the ^{22}Na photopeak (Fig. 23). The dataset can be further refined *a posteriori* by exploiting the energy information from the NaI detector. As shown in Sec. 3, the 511 keV photopeak was previously identified in the NaI spectrum. By retaining only events whose NaI signal falls within a narrow window around the fitted photopeak (the chosen window was [1300,1340] ADC, centred around the photopeak at 1320 ADC), accidental coincidences and events in which the NaI detector registers only partial energy deposition (for example through Compton scattering) are largely rejected. The resulting spectra are shown in Fig. 23. The upper panel displays the NaI spectrum; the lower panel compares the reconstructed LYSO cluster spectra before and after the NaI energy cut.

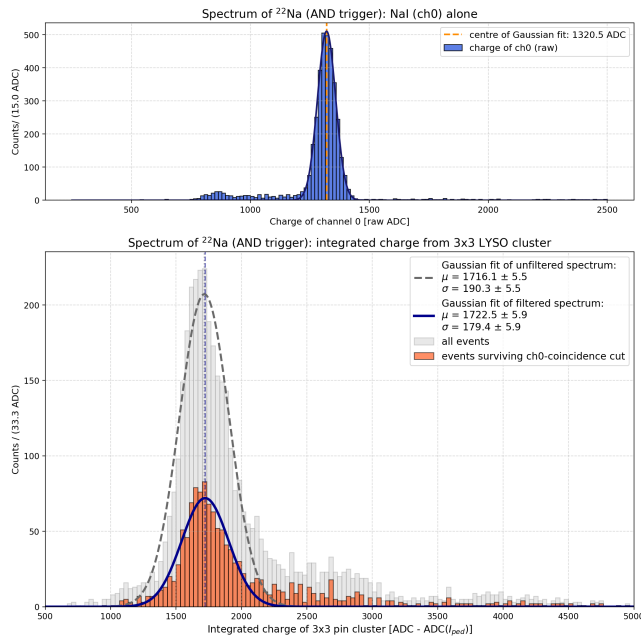


Figure 23: Spectrum of ^{22}Na via the back-to-back coincidence method. Upper panel: NaI spectrum with the fitted photopeak. Lower panel: integrated charge of the reconstructed 3×3 LYSO cluster before and after applying the NaI energy cut. Data acquired during a 100-minute run at $\text{HV} = 980 \text{ V}$.

The doubly-filtered LYSO photopeak was fitted with a Gaussian function, yielding a centre of $(1722 \pm 6) \text{ ADC}$.

⁵Or any other source which emits back-to-back.

This value is consistent with the ^{22}Na peak position obtained previously in Sec. 4.2 within 1σ .

4.4 Gain-HV dependence

Triggering on the AND coincidence between the LYSO and NaI detector signals, as explained in the previous section (Sec. 4.3), the reconstructed 3×3 cluster peak position of the ^{22}Na source's 511 keV peak was obtained at varying values of the supply HV⁶, assuming a uniform gain across all pins of the cluster. Peak positions are plotted against supply voltage in Figure 24, and fitted with an exponential function $y = Ae^{bx}$, whose parameters are in Table 11.

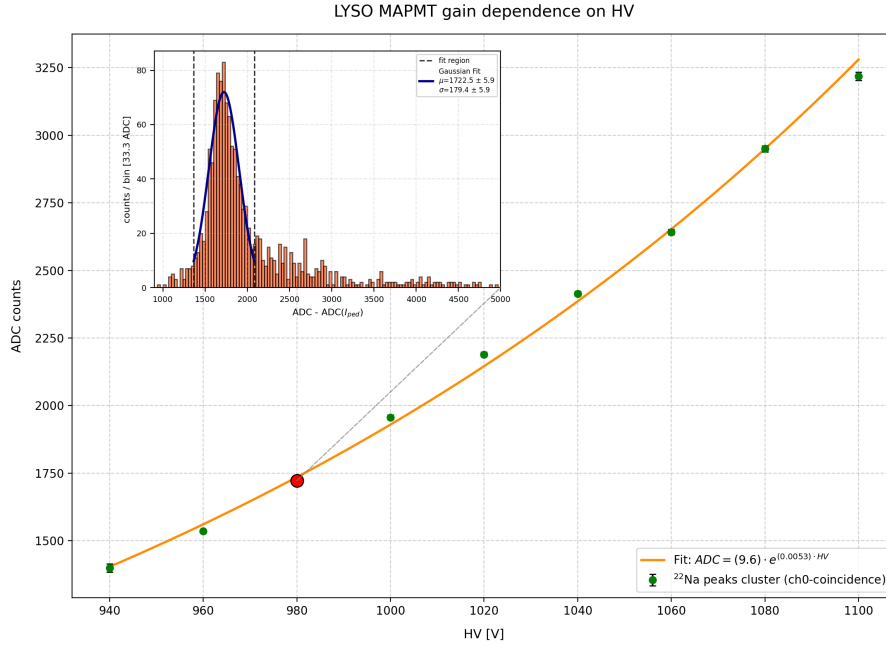


Figure 24: Position of the ^{22}Na reconstructed cluster peak as a function of applied HV. For each HV value the peak position and its uncertainty were extracted via Gaussian fit, as illustrated in the inset for the example case $\text{HV} = 980\text{ V}$. The data points of the main plot are fitted with an exponential function $y = Ae^{bx}$.

A [ADC]	b [V^{-1}]
10 ± 1	0.0053 ± 0.0001

Table 11: Parameters extracted from the exponential fit to the LYSO gain-HV curve.

An operating HV of 980 V was selected. As in the NaI case, this is roughly the highest value such that the available sources don't saturate the QDC dynamic range, the limiting factor being the ^{60}Co peaks.

4.5 Background suppression via charge-sharing topology

In this subsection we first characterise the general features of the charge-sharing topology across LYSO pins for single events, and then investigate whether this information can be used to construct a background suppression filter.

It is important to clarify at the outset what these topological methods can and cannot do. They can reject accidental background events (cases in which two unrelated photons happen to arrive within the same acquisition gate

⁶Pedestal values for relevant channels are reported in Appendix A.

and produce an atypical charge-sharing pattern). They cannot, however, distinguish a genuine photon from the radioactive source from a genuine ^{176}Lu background photon of similar energy, since both produce indistinguishable signals in the detector. These methods therefore reduce accidental contamination but do not provide a clean separation of source against background.

The following quantities are used throughout, when referring to a single event:

- Q_{central} , the pedestal-subtracted charge integrated by the central pin alone (ie pin 36);
- $Q_{\text{neighbours}}$, the pedestal-subtracted charge integrated by the central pin's eight neighbours;
- $Q_{\text{tot}} = Q_{\text{central}} + Q_{\text{neighbours}}$, the total pedestal-subtracted charge of the 3×3 cluster;
- $R = Q_{\text{central}}/Q_{\text{tot}}$, the fraction of the total cluster charge localised in the central pin.

Figure 25 shows the two-dimensional correlation between Q_{central} and $Q_{\text{neighbours}}$ in the 9 hour single-trigger ^{22}Na run (Tab. 8). A *prima facie* identical figure is obtained from any of the other single-trigger runs and from the background run itself, as the signal-to-background ratio is low in all of them⁷.

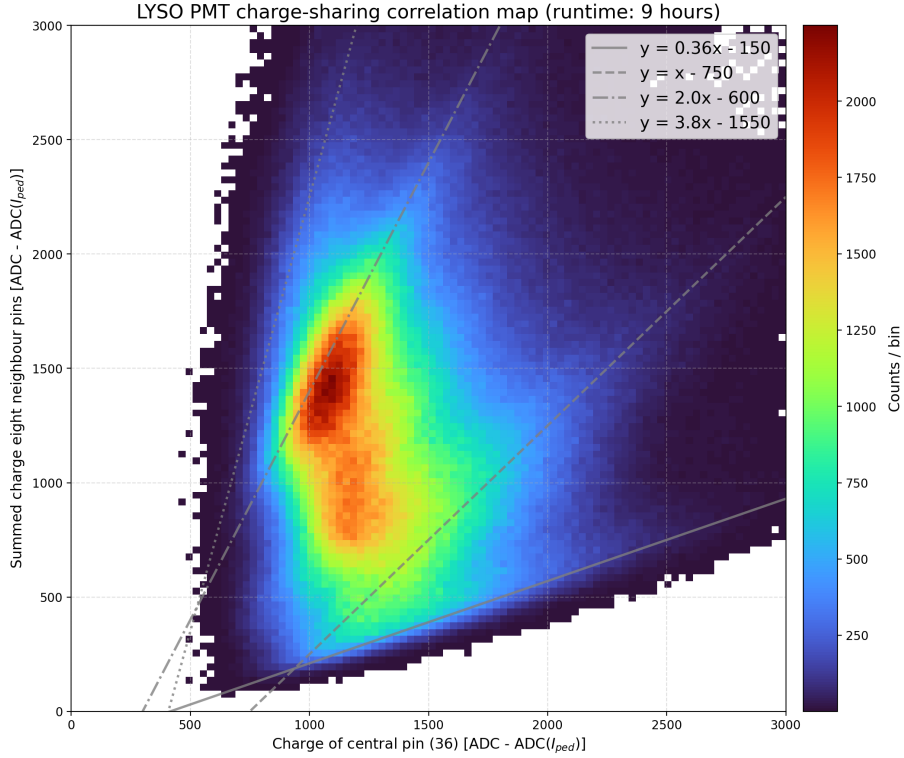


Figure 25: Correlation between Q_{central} (horizontal axis) and $Q_{\text{neighbours}}$ (vertical axis) in the 9 hour single-trigger ^{22}Na run. Aside from the dense central region corresponding to the "peak" of the background spectrum, several stripes are also visible. They are highlighted with grey lines and commented on in the text.

Qualitatively, the four visible bands can be interpreted as distinct charge-sharing scenarios within the LYSO matrix. Each band approximately satisfies:

$$Q_{\text{neighbours}} \approx m Q_{\text{central}} \implies Q_{\text{tot}} \approx (m+1) Q_{\text{central}},$$

so their slopes m roughly correspond to $Q_{\text{tot}}/Q_{\text{central}} - 1 = 1/R - 1$.

⁷The signal-to-background ratio can be estimated to be around 5% for these runs by integrating the histograms in Figures 18, 19, 20.

Physically, a lower slope corresponds to an interaction that occurred near the centre of the central pin, while a higher slope suggests an interaction near a pin boundary (either on the edge between two pins, or in the corner between three or four), where charge is shared more with neighbouring pins.

The bands identified in Fig. 25 are:

- “Band 1” ($m \approx 0.36$) where the charge contained in the neighbouring pins is about a third of that in the central pin;
- “Band 2” ($m \approx 1$) where the charge integrated in the central pin roughly balances that integrated in its 8 surrounding neighbours;
- “Band 3” ($m \approx 2$) where the neighbouring pins contain roughly twice the charge in the central pin;
- “Band 4” ($m \approx 3.8$) where charge is even more shared between pins, and an even lesser share of the total charge is in the central pin.

Having characterised the charge-sharing topology, we now investigate whether it can be used as a background suppression filter. The strategy is to use the background-subtracted 2D correlation as a reference, and then restrict the single-trigger spectrum to events whose $(Q_{\text{central}}, Q_{\text{neighbours}})$ pair falls within a region identified from that reference.

Similarly to the 1D background subtraction performed in Sec. 4.2, the rescaled background 2D histogram is subtracted from the single-trigger ^{22}Na 2D histogram. The result is shown in Fig. 26.

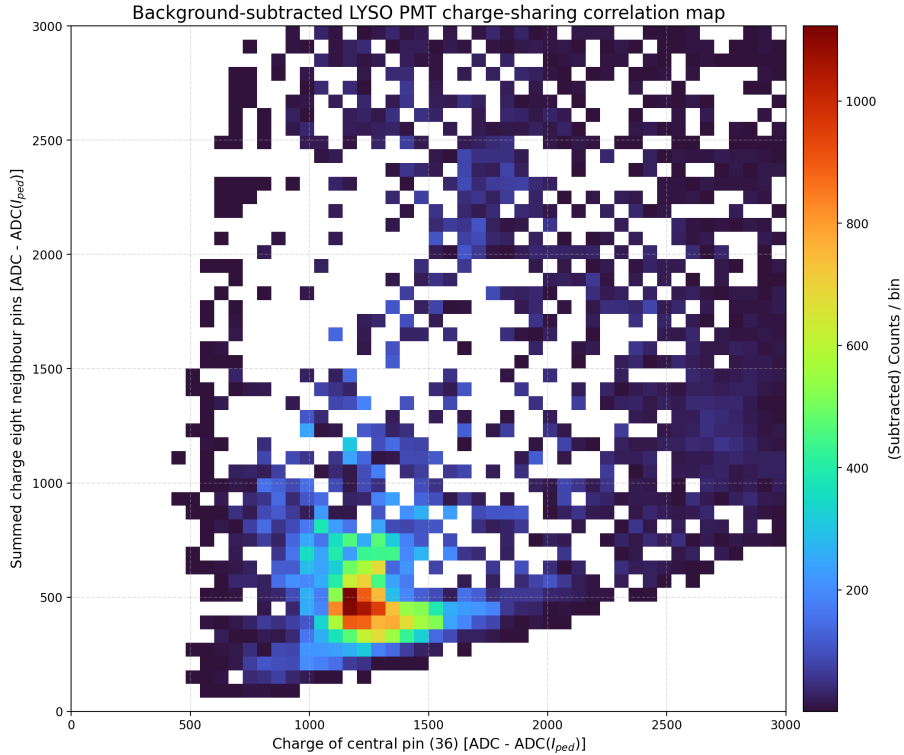


Figure 26: Background-subtracted 2D histogram of Q_{central} vs. $Q_{\text{neighbours}}$ for the ^{22}Na single-trigger run. The region of events corresponding to genuine ^{22}Na interactions is visible above the subtracted background. Bins containing negative counts after subtraction are displayed as white, and would ideally be removed by repeating the measurements for much longer runtimes to increase event statistics.

A filter is then applied to the single-trigger ^{22}Na 1D spectrum by accepting only events whose $(Q_{\text{central}}, Q_{\text{neighbours}})$

coordinates fall within a region centred on the excess visible in Fig. 26. The filtered region and the resulting spectrum are shown in Fig. 27.

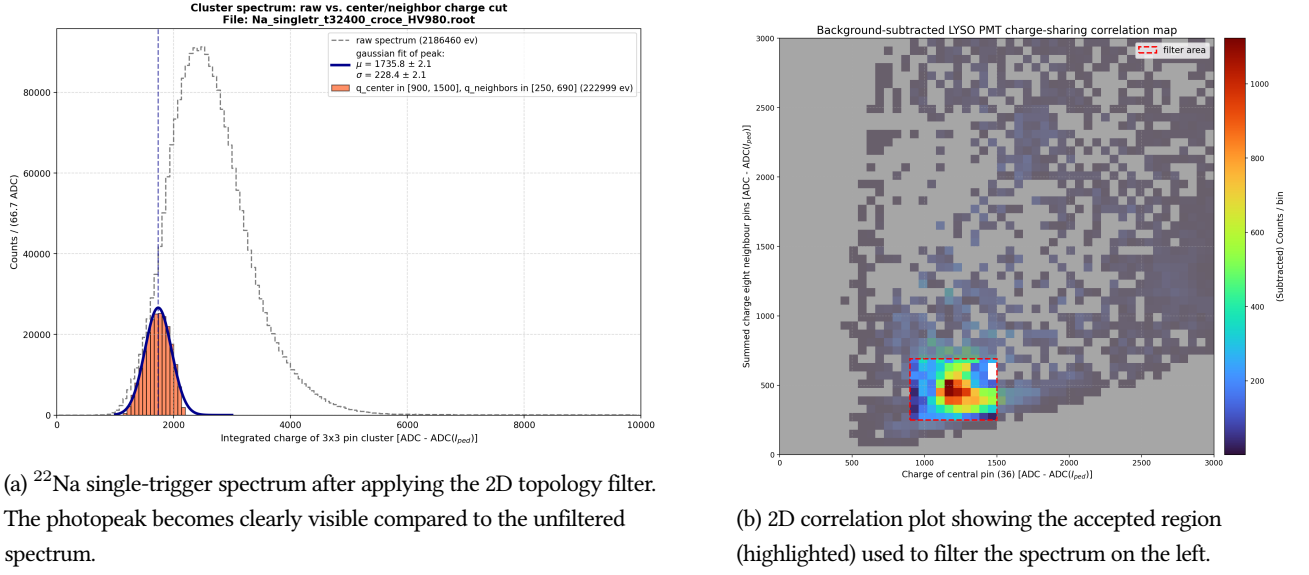


Figure 27: Application of the charge-sharing topology filter to the ^{22}Na single-trigger dataset. Events outside the highlighted region in the 2D plot (right) are rejected; the resulting 1D spectrum (left) shows an improved signal-to-background ratio compared to the unfiltered case.

The resulting peak (1736 ± 11) ADC is compatible within 1σ with the value (1722 ± 13) ADC obtained from Sec. 4.3 (where the total uncertainty is evaluated with Eq. 3.2).

This method has several limitations worth noting: the filter region is chosen somewhat arbitrarily and is specific to a given source and run configuration; it requires high statistics to construct a reliable background-subtracted 2D reference (which is why it was only applied to the ^{22}Na dataset here); and it is most naturally suited to *Configuration A*, where TDC information is not recorded.

4.6 Background suppression via timing

We now investigate whether timing information from the can be exploited to suppress background for a generic radioactive source, where no external coincidence condition is applicable (as in the case of ^{22}Na). As with the topological filter above, this method can only reduce accidental contamination, it cannot separate genuine source photons from genuine ^{176}Lu background photons of similar energy.

A first point to make is that timing information *alone* is not sufficient to fully resolve a source's peak against the LYSO background, and can at most be a supplemental filter to be used with other methods (e.g. the ones mentioned above). The reason is straightforward and is illustrated in Fig. 29: in the vast majority of events ($\approx 80\%$), only a single TDC channel fires (the trigger pin, channel 1). Even a maximally aggressive timing cut therefore acts on at most $\approx 20\%$ of events.

The experimental setup corresponds to *Configuration B* (Fig. 15b). The added TDC is operated in COMMON START mode, with the QDC gate providing the common start signal. The signals from the five selected LYSO channels and the NaI channel are delayed⁸ so that pulses arriving within the TDC acquisition window of about 152 ns⁹

⁸The NaI signal was delayed by 100 ns, while the LYSO signals were delayed by 150 ns, in addition to the intrinsic delays introduced by the electronics and LEMO cables (approximately 5 ns/m).

⁹The acquisition window is set by the Full Scale Range (FSR) register, configured to a value of 240 in the DAQ software. According to the CAEN V775N manual [7], this corresponds to a Least Significant Bit of $8.9/240 \approx 37\text{ps}$, giving a maximum measurable range of

have their arrival times recorded.

Figure 28 shows the raw TDC distributions for the six connected channels in a typical single-trigger run. Channel 1 (LYSO pin 36) exhibits an almost delta-function-like distribution because it is the trigger channel: its recorded time reflects the fixed delay between the discriminator output that generates the trigger and the common-start signal reaching the TDC. The remaining channels display broader distributions.

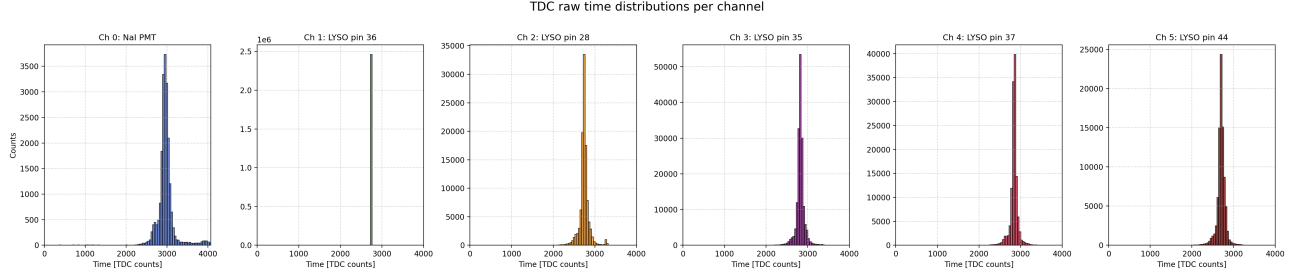


Figure 28: Raw TDC distributions for the six connected channels during a typical run (single-trigger ^{22}Na , runtime: 100 min). Events with value -999 , corresponding to no signal detected within the TDC acquisition window, have been removed for clarity; these constitute the majority of entries for all channels except the trigger channel. It should be emphasised that the raw TDC values for individual channels do not directly represent physical arrival times. Each channel is affected by fixed offsets arising from differences in cable lengths, electronic modules and signal propagation delays. Hence, a smaller raw TDC value does not necessarily correspond to a signal that physically arrived earlier.

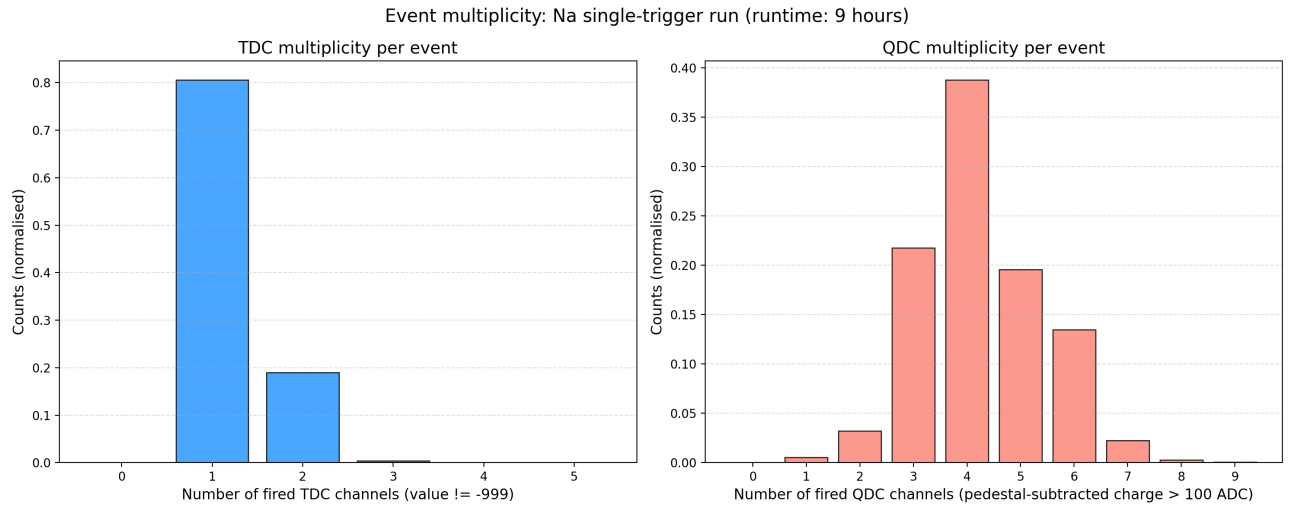


Figure 29: (Left) TDC hit multiplicity in the ^{22}Na single-trigger run, defined as the number of channels (out of five) measuring a value $\neq -999$. The vast majority of events have only one TDC hit (the trigger channel), and the fraction with more than two hits is negligible. (Right) QDC hit multiplicity, defined as the number of pins (out of nine) measuring a pedestal-subtracted charge above 100 ADC. Both distributions are essentially identical for the other single-trigger runs (^{60}Co , ^{137}Cs) and for the background run, as each has a similarly low signal-to-background ratio.

Nevertheless, we can still apply a timing cut as an *a posteriori* filter to reject events in which multiple pins fire at mutually inconsistent times within the same gate. The physical motivation is:

- if two or more TDC channels fire within a narrow time window of each other, they can reasonably be assumed to arise from the same photon interacting in the crystal;

$4096 \times 37\text{ps} \approx 152\text{ns}$ per channel. Any signal arriving after this window is recorded as an overflow and assigned the placeholder value -999 .

- if two or more channels fire at significantly different times within the gate, they are likely to arise from distinct, unrelated photons arriving in the same gate, and the event should be rejected.

In practice, since channel 1 (the trigger pin) has an essentially constant TDC value (Fig. 28), we implement the cut by requiring that any non-trigger channel recording a hit satisfies:

$$|TDC_i - TDC_1| < \text{threshold}_i(\sigma_i), \quad i \in \{2, 3, 4, 5\},$$

where σ_i is the standard deviation of the TDC_i distribution. Events for which any firing non-trigger channel fails this condition are discarded. Events in which a given non-trigger pin records no hit (-999) are not affected by this cut.

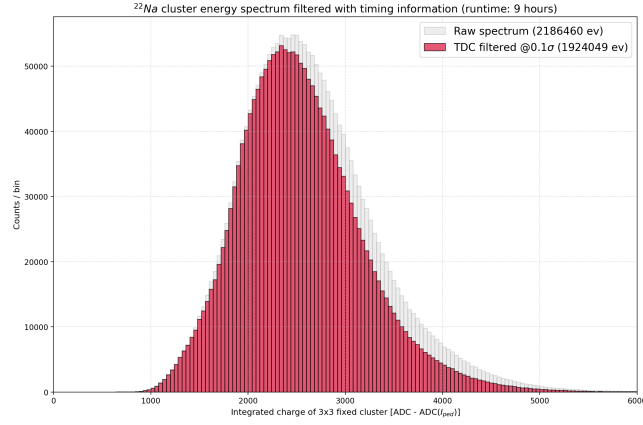


Figure 30: LYSO spectrum before and after the TDC timing cut, with the threshold set to $0.1\sigma_i \forall i$. This deliberately strict threshold was chosen to illustrate that even rejecting all events whose non-trigger TDC hits deviate by more than 0.1σ leaves the spectrum basically unchanged.

Although physically motivated, this filter leaves the spectrum largely unchanged for the reasons specified above (it acts on a minority of events). This is clearly evident in Fig. 30, where the (quite radical) filter threshold $0.1\sigma_i \forall i$ was imposed.

As expected, we conclude that this timing condition alone is insufficient to address the background-suppression problem for a generic source. It may, however, provide a useful supplemental rejection when combined with the topological filter described above.

5 Conclusions

This work aimed to characterise both the NaI reference scintillator and the 64-channel LYSO scintillator and to develop strategies for identifying genuine signals against the intrinsic ^{176}Lu radioactivity of LYSO, which produces a background that dominates the acquired spectra in all single-trigger runs.

Summary of results

The NaI detector was fully characterised. the operating voltage was set to 550 V, within the manufacturer-specified range of 500 V to 800 V. Its gain-HV dependence was measured and found to follow an exponential trend in the 500 V to 600 V range explored, and an ADC-to-energy calibration was obtained using three radioactive sources (^{137}Cs , ^{60}Co , ^{22}Na), yielding a linear calibration with slope $m = (2.52 \pm 0.01)$ ADC/keV. The relative energy resolution σ/E was found to decrease with energy from $\sim 3.3\%$ at 511 keV to $\sim 2.1\%$ at 1332 keV, consistently with the expected $1/\sqrt{E}$ scaling. All five photopeaks used for the calibration were resolved individually, confirming the NaI detector's suitability as a reference system.

For the LYSO detector, the operating voltage was set to 980 V, and an ADC-to-energy calibration was obtained with slope $m = (2.6 \pm 0.1)$ ADC/keV. The larger relative uncertainty compared to NaI reflects the smaller number of calibration points and the fact that the two ^{60}Co gamma lines at 1173 and 1332 keV could not be resolved, requiring their average to be used as a single calibration point. The energy resolution of the LYSO detector was found to be significantly worse than that of NaI over the same energy range, $\sigma/E \simeq 14.8\%$ at 511 keV and $\sigma/E \simeq 14.1\%$ at 661.7 keV, consistent with the lower light yield per channel and the intrinsic background. The spatial distribution of scintillation light was analysed: it was found that for a typical event, integrating a 3×3 pin cluster centred on the pin with the largest charge provides a reliable estimator of the total event energy. Four distinct charge-sharing bands were identified in the $Q_{\text{central}} - Q_{\text{neighbours}}$ correlation plane, which were interpreted as interactions occurring at different positions relative to the pin boundaries.

Three complementary background suppression strategies were investigated:

1. **Back-to-back coincidence (AND trigger).** For the ^{22}Na source, requiring a hardware coincidence between the LYSO and NaI detectors, combined with an energy cut on the NaI signal, provided by far the most effective suppression: the ^{22}Na 511 keV photopeak was cleanly resolved with no additional background subtraction required. The doubly-filtered peak was found at (1722 ± 6) ADC, consistent within 1σ with the value obtained from the standard background-subtraction method.
2. **Charge-sharing topology filter.** By restricting accepted events to those whose $(Q_{\text{central}}, Q_{\text{neighbours}})$ pair falls within a region selected in the background-subtracted 2D correlation, the ^{22}Na photopeak was resolved from the single-trigger spectrum without an external coincidence detector. The method is demonstrably effective but ad hoc: the filter region must be chosen for each source and requires high-statistics background runs for good subtraction, which limited its application here to the ^{22}Na dataset alone.
3. **TDC timing cut.** Requiring that any non-trigger TDC channel recording a hit lies within a narrow time window of the trigger channel was found to leave the spectrum essentially unchanged, even at the deliberately radical threshold of $0.1\sigma_i$. This is a direct consequence of the low TDC multiplicity: in approximately 80% of events only a single TDC channel fires, leaving no timing information on which to cut. The method is therefore insufficient on its own, though it could be combined with the topological filter.

It is important to note that methods (2) and (3) are fundamentally limited in what they can achieve: they can only reject *accidental* contamination (cases where two unrelated photons arrive within the same acquisition gate and

produce an atypical pattern), but cannot distinguish a genuine photon from the radioactive source from a genuine ^{176}Lu background photon of the same energy, since both produce identical signals in the detector.

5.1 Proposals for future experiments

Several improvements would extend and strengthen the results obtained here.

- **Adding more TDC channels.** The most significant hardware constraint throughout this work was the limited number of discriminator channels (7 available) and TDC inputs (16 available), which forced us to read out only a subset of the 64 LYSO pins and to use a single-pin trigger in *Configuration B*. Connecting more pins to the TDC would increase the TDC hit multiplicity per event, making the timing cut in Section 4.6 substantially more powerful.
- **Acquiring higher statistics runs for the 2D topology filter.** The charge-sharing topology method (Section 4.5) was limited to the ^{22}Na dataset because the background-subtracted 2D histogram requires high statistics to identify the signal region reliably: with 9-hour signal and background runs, the subtracted 2D map is still noisy and the filter region must be chosen by eye. Longer background runs would allow a more principled and reproducible definition of the filter region, and would make it applicable to the ^{137}Cs and ^{60}Co sources as well.
- **Measuring the gain-HV dependence independently for each pin.** Throughout this work, a uniform gain was assumed across all LYSO pins. In practice, individual anodes of an multi-anode PMT can differ in gain, and this variation is itself HV-dependent. This assumption introduces a systematic uncertainty in the reconstructed cluster charge that is difficult to quantify without specific per-pin measurements. A straightforward improvement would be to measure the gain-HV curve (as done in Section 4.4 for the cluster as a whole) for each individual pin, for example using the ^{137}Cs photopeak as a reference, and derive individual calibration factors. This would also allow the 3×3 cluster sum to be corrected for inter-channel gain variations before fitting, potentially improving the energy resolution.
- **Combining the topology and timing filters.** As noted at the end of Section 4.6 and Section 4.5, neither the charge-sharing filter nor the TDC timing cut is sufficient on its own for generic sources. Their combination (accepting only events that pass both the 2D topology region requirement and the multi-channel timing consistency condition) has not been tested here, but is a natural next step.

A Appendix

The pedestal values for the 32 channels of the QDC module were measured on three separate occasions, corresponding to different data-taking sessions. The acquisition date is reported in the caption of each table (Table 12, Table 13, Table 14). For each channel, the pedestal value and its uncertainty were extracted from a Gaussian fit to the corresponding ADC counts spectrum. The uncertainty is the statistical uncertainty on the fitted mean.

Ch	Pedestal [ADC]	Ch	Pedestal [ADC]
0	124.4029 ± 0.0006	16	445.6726 ± 0.0003
1	446.4691 ± 0.0003	17	474.6053 ± 0.0004
2	453.6380 ± 0.0004	18	408.9379 ± 0.0003
3	403.2420 ± 0.0004	19	457.9262 ± 0.0004
4	427.2941 ± 0.0004	20	436.9149 ± 0.0003
5	419.1673 ± 0.0007	21	461.9966 ± 0.0004
6	450.5580 ± 0.0003	22	432.0763 ± 0.0003
7	421.3103 ± 0.0004	23	412.5315 ± 0.0005
8	438.3920 ± 0.0003	24	409.0104 ± 0.0004
9	441.4552 ± 0.0004	25	407.5138 ± 0.0003
10	410.5981 ± 0.0003	26	437.5038 ± 0.0003
11	465.5779 ± 0.0004	27	439.4948 ± 0.0003
12	426.4848 ± 0.0004	28	418.8417 ± 0.0003
13	435.3054 ± 0.0004	29	459.2080 ± 0.0004
14	437.8687 ± 0.0004	30	424.8704 ± 0.0003
15	391.0810 ± 0.0005	31	391.9038 ± 0.0004

Table 12: Measured pedestal values in ADC counts for 32 channels of the QDC module (5^{th} May 2026).

Ch	Pedestal [ADC]	Ch	Pedestal [ADC]
0	114.25 ± 0.01	16	254.702 ± 0.004
1	183.34 ± 0.03	17	264.879 ± 0.004
2	200.04 ± 0.01	18	222.037 ± 0.003
3	175.16 ± 0.02	19	253.194 ± 0.004
4	199.74 ± 0.01	20	251.232 ± 0.004
5	200.52 ± 0.02	21	252.796 ± 0.004
6	254.88 ± 0.01	22	238.104 ± 0.004
7	240.16 ± 0.02	23	230.320 ± 0.004
8	258.35 ± 0.01	24	232.576 ± 0.004
9	251.21 ± 0.02	25	222.760 ± 0.004
10	226.25 ± 0.02	26	238.591 ± 0.004
11	254.734 ± 0.004	27	236.060 ± 0.003
12	237.791 ± 0.003	28	236.621 ± 0.004
13	245.168 ± 0.003	29	266.469 ± 0.004
14	247.698 ± 0.004	30	221.948 ± 0.003
15	221.434 ± 0.004	31	217.039 ± 0.003

Table 13: Measured pedestal values in ADC counts for 32 channels of the QDC module (6^{th} July 2026).

Ch	Pedestal [ADC]	Ch	Pedestal [ADC]
0	114.12 ± 0.01	16	255.322 ± 0.004
1	185.69 ± 0.03	17	265.951 ± 0.003
2	201.54 ± 0.01	18	223.027 ± 0.002
3	176.64 ± 0.02	19	254.404 ± 0.004
4	201.51 ± 0.01	20	252.108 ± 0.003
5	202.52 ± 0.02	21	254.019 ± 0.003
6	255.64 ± 0.01	22	239.129 ± 0.004
7	241.12 ± 0.02	23	231.487 ± 0.004
8	259.02 ± 0.01	24	233.842 ± 0.004
9	252.07 ± 0.03	25	223.642 ± 0.004
10	227.469 ± 0.004	26	239.515 ± 0.004
11	255.483 ± 0.004	27	237.082 ± 0.003
12	239.618 ± 0.004	28	237.226 ± 0.004
13	246.449 ± 0.004	29	267.826 ± 0.004
14	248.744 ± 0.004	30	222.910 ± 0.003
15	222.355 ± 0.004	31	218.005 ± 0.003

Table 14: Measured pedestal values in ADC counts for 32 channels of the QDC module (16th July 2026).

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